SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE CODE: ITA 06 **COURSE NAME: MACHINE LEARNING**

COURSE OUTCOME COVERED

CO3: Bayes Theorem – Concept Learning – Maximum Likelihood – Minimum Description Length Principle – Bayes Optimal Classifier – Gibbs Algorithm – Naïve Bayes Classifier – Bayesian Belief Network – EM Algorithm – Probability Learning – Sample Complexity – Finite and Infinite Hypothesis Spaces – Mistake Bound Model, (BL 6)

Assessment Tool Weightage: 20%

QUESTIONS: 15 **Total Marks:25**

Annexure A

Case Study

Code of Conduct:

I _M. Pranay (192525382) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: *M. Pranay*

Annexure A

Short Answer Test

1, A hospital aims to develop an intelligent system to diagnose diseases based on patient symptoms and medical history. Using concepts such as Bayes' Theorem, Naïve Bayes classifiers, and Bayesian Belief Networks, design a probabilistic model for diagnosis. Analyze how uncertainty is handled and how conditional dependencies between symptoms influence predictions. Discuss the role of probability learning and evaluate the effectiveness of your approach with respect to real-world constraints.

2. An organization wants to build a spam detection system for emails using machine learning techniques. Using Naïve Bayes and Bayes Optimal Classifier concepts, design a classification system. Compare Maximum Likelihood estimation with Minimum Description Length (MDL) principle in feature selection. Analyze system performance, discuss trade-offs in model complexity, and evaluate how sample size and hypothesis space affect classification accuracy.

3. A retail company wants to segment its customers based on purchasing behavior, but the data contains hidden patterns and incomplete information. Apply the EM algorithm to identify clusters and estimate underlying distributions. Analyze convergence behavior, initialization challenges, and the role of probability learning. Discuss how the results can be used for targeted marketing and evaluate limitations of the approach.

4. A research team is developing a concept learning system to classify medical images into normal and abnormal categories. Using hypothesis space search and the Candidate Elimination approach, design a model that learns from labeled examples. Analyze the impact of finite vs infinite hypothesis spaces, sample complexity, and mistake bound model on learning performance. Evaluate how well the system generalizes to unseen data.

5. A financial institution needs to make investment decisions under uncertain market conditions. Using Bayesian inference, Gibbs algorithm, and probability learning, design a system to predict outcomes and update beliefs dynamically. Analyze how prior knowledge influences predictions, evaluate risk using posterior probabilities, and discuss how the model adapts as new data becomes available. Consider computational challenges and optimization strategies.

6. An autonomous vehicle system must make real-time driving decisions under uncertain environmental conditions such as weather, traffic, and pedestrian movement. Using Bayesian inference, Naïve Bayes classifiers, and probabilistic reasoning, design a model to predict safe driving actions. Analyze how uncertainty and sensor noise are handled, and evaluate system reliability under dynamic conditions.

7. A healthcare organization aims to predict patient risk levels for chronic diseases using historical and lifestyle data. Apply probability learning, Bayes' Theorem, and hypothesis space search to design the model. Analyze the impact of prior probabilities, feature dependencies, and sample size. Evaluate the model's interpretability and its usefulness in preventive healthcare strategies.

8. A bank wants to detect fraudulent transactions using machine learning under uncertain and evolving patterns. Using Bayesian Belief Networks and EM algorithm concepts, design a detection system. Analyze hidden variables, incomplete data handling, and convergence issues. Evaluate how probability-based reasoning improves detection accuracy and reduces false positives in real-world scenarios.

9. An e-commerce platform aims to recommend products to users based on browsing and purchase history. Using Naïve Bayes and Bayesian learning concepts, design a recommendation model. Analyze how conditional probabilities influence recommendations and how uncertainty in user preferences is managed. Evaluate scalability and performance in handling large, dynamic datasets.

10. A research institute is building a system to predict climate changes using historical environmental data. Apply probabilistic models such as Bayesian inference and EM algorithm to handle incomplete and noisy datasets. Analyze convergence behavior, uncertainty modeling, and the influence of prior knowledge. Evaluate the system's effectiveness in long-term environmental forecasting.

11. A telecommunications company aims to predict customer churn using historical usage data and service feedback. Using probabilistic models such as **Naïve Bayes**, **Bayesian inference**, and hypothesis space search, design a predictive system. Analyze how prior probabilities and feature dependencies influence churn prediction. Evaluate how uncertainty in customer behavior is handled and discuss the effectiveness of the model in improving customer retention strategies.

12. A smart healthcare system is designed to monitor patients in real-time using wearable sensors. Using **Bayesian Belief Networks** and probability learning, develop a model to detect anomalies in vital signs. Analyze how conditional dependencies between physiological parameters are modeled and how uncertainty due to sensor noise is handled. Evaluate system performance and discuss challenges in real-time deployment.

13. A logistics company wants to optimize delivery routes under uncertain traffic and weather conditions. Using **Bayesian inference** and probabilistic reasoning, design a system that predicts optimal routes dynamically. Analyze how prior knowledge and real-time data updates influence decision-making. Evaluate computational complexity and discuss how uncertainty impacts route optimization efficiency.

14. A cybersecurity system is developed to detect network intrusions using probabilistic learning methods. Apply **Naïve Bayes classifiers**, **EM algorithm**, and Bayesian learning to model normal and abnormal network behavior. Analyze how hidden patterns and incomplete data are handled. Evaluate the system's effectiveness in reducing false alarms and improving detection accuracy in dynamic environments.

15. An agriculture-based system aims to predict crop yield under uncertain environmental conditions such as rainfall, soil quality, and temperature. Using **Bayesian inference** and probability learning, design a predictive model. Analyze how prior agricultural knowledge and seasonal variations influence predictions. Evaluate the model's reliability and discuss its impact on decision-making for farmers and policy planning.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	**30**	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	**20**	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	**20**	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	**15**	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	**15**	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

ANSWERS

1. Hospital Disease Diagnosis Using Bayesian Methods

Introduction:

A hospital can develop an intelligent disease diagnosis system using Bayes' Theorem, Naïve Bayes classifiers and Bayesian Belief Networks.

Approach:

1. Collect patient data such as symptoms, age and medical history.
2. Apply Bayes' Theorem to calculate the probability of a disease.
3. Use the Naïve Bayes classifier to predict diseases based on symptoms.
4. Construct a Bayesian Belief Network to represent relationships between symptoms and diseases.
5. Calculate posterior probabilities based on patient information.
6. Predict the most probable disease.

Handling Uncertainty:

- Probability helps make decisions when symptoms are incomplete.
- Conditional dependencies show relationships between symptoms.
- Probability learning improves predictions using previous patient records.
- New patient information helps update predictions.

Real-World Constraints:

- Patient data must be accurate and secure.
- Missing medical information may affect predictions.
- The system should provide quick and reliable results.

Conclusion:

Bayesian methods help hospitals predict diseases effectively by handling uncertainty and analysing relationships between symptoms.

2. Spam Detection System Using Naïve Bayes

Introduction:

An organization can develop a spam detection system using Naïve Bayes and Bayes Optimal Classifier concepts to classify emails as spam or not spam.

Approach:

1. Collect a dataset containing spam and normal emails.
2. Preprocess the email text and extract important words.
3. Apply the Naïve Bayes classifier.
4. Calculate the probability of an email being spam.
5. Use the Bayes Optimal Classifier to select the most probable classification.
6. Evaluate the system using accuracy and precision.

MLE and MDL Comparison:

- Maximum Likelihood Estimation (MLE): Estimates probabilities using the available training data.

- Minimum Description Length (MDL): Selects a model by balancing model complexity and data representation.

Analysis:

- A larger dataset generally improves classification accuracy.
- A small dataset may produce incorrect predictions.
- A large hypothesis space increases model complexity.
- Simple models are faster, while complex models may capture more patterns.
- Proper feature selection improves spam detection.

Conclusion:

Naïve Bayes and Bayes Optimal Classifier techniques help identify spam emails efficiently and improve email security.

3. Customer Segmentation Using EM Algorithm

Introduction:

A retail company can use the Expectation-Maximization (EM) algorithm to segment customers based on their purchasing behaviour and identify hidden patterns.

Approach:

1. Collect customer purchase history and transaction data.
2. Preprocess the data and handle missing values.
3. Select the number of customer clusters.
4. Initialize the model parameters.
5. Perform the Expectation (E) step to calculate cluster probabilities.
6. Perform the Maximization (M) step to update parameters.
7. Repeat both steps until the algorithm converges.
8. Analyse the final customer groups.

Analysis:

- EM identifies hidden patterns in customer behaviour.
- Different initial values may produce different clusters.
- The algorithm may converge to a local optimum.
- Probability learning helps estimate customer group distributions.
- Incomplete information can affect clustering results.

Applications in Retail:

- Identify frequent customers.
- Provide personalized offers.
- Improve targeted marketing.
- Understand customer purchasing patterns.

Conclusion:

The EM algorithm helps retail companies identify customer segments and improve marketing strategies through purchasing behaviour analysis.

4. Medical Image Classification Using Concept Learning

Introduction:

A research team can develop a concept learning system to classify medical images into normal and abnormal categories using hypothesis space search and the Candidate Elimination algorithm.

Approach:

1. Collect labelled medical images.
2. Extract important image features.
3. Define the hypothesis space.
4. Initialize the Specific (S) and General (G) boundaries.
5. Train the model using positive and negative examples.
6. Update the hypothesis boundaries based on training examples.
7. Classify new images as normal or abnormal.

Analysis:

- Hypothesis space contains possible classification rules.
- Finite hypothesis spaces contain a limited number of hypotheses.
- Infinite hypothesis spaces contain unlimited possible hypotheses.
- Sample complexity determines the number of examples required for learning.
- The mistake bound model measures the maximum number of prediction errors.
- More training examples can improve generalization.

Generalization:

The trained system can classify unseen medical images based on learned patterns. Proper training examples improve classification performance.

Conclusion:

Concept learning and Candidate Elimination help classify medical images by learning rules from labelled examples.

5. Investment Prediction Using Bayesian Inference

Introduction:

A financial institution can develop an investment prediction system using Bayesian inference, Gibbs algorithm and probability learning to make decisions under uncertain market conditions.

Approach:

1. Collect historical stock prices and market information.
2. Identify important factors such as market trends and investment risks.
3. Apply Bayesian inference to calculate the probability of investment outcomes.
4. Use prior knowledge to estimate initial probabilities.
5. Apply the Gibbs algorithm to generate samples from complex probability distributions.
6. Calculate posterior probabilities.
7. Update predictions when new market data becomes available.

Analysis:

- Prior knowledge influences initial predictions.
- Posterior probabilities help estimate investment risks.

- Probability learning improves predictions using new information.
- Gibbs sampling helps perform probabilistic inference.
- Continuous data updates help the system adapt to changing market conditions.

Challenges:

- Large datasets require more computation.
- Market conditions are highly unpredictable.
- Complex probability calculations may take more time.

Optimization Strategies:

- Use efficient sampling methods.
- Update the model regularly.
- Use suitable prior probabilities.
- Reduce unnecessary calculations.

Conclusion:

Bayesian inference and the Gibbs algorithm help financial institutions analyse uncertainty, estimate investment risks and make better investment decisions.